Title

Computational immunophenotyping of biomaterial-induced foreign body response reveals macrophage and fibroblast programs associated with fibrotic remodelling

Background

Implanted biomaterials trigger a foreign body response (FBR) characterized by immune cell recruitment, persistent inflammation, and, in many cases, fibrotic encapsulation of the implant. Macrophages play a central role in orchestrating this response by integrating material-derived cues and coordinating downstream inflammatory and remodelling processes. Fibroblasts act as key stromal effectors, responding to immune-derived signals by producing extracellular matrix (ECM) components that ultimately define the extent and quality of fibrotic tissue formation. Although both cell types are well recognized as critical regulators of FBR, the transcriptional diversity of macrophages and fibroblasts, as well as the immune–stromal communication programs operating at biomaterial interfaces, remain incompletely understood (Yang *et al.,* 2023).

Single-cell RNA sequencing (scRNA-seq) enables unbiased characterization of cellular composition and functional states within complex tissues. In the context of biomaterial implantation, this approach provides an opportunity to resolve immune and stromal heterogeneity and to link specific cellular programs with inflammatory or fibrotic outcomes (Liang et al., 2024). Here, we apply computational immunophenotyping to publicly available scRNA-seq data from biomaterial implantation sites to characterize macrophage and fibroblast states associated with fibrotic remodelling and to explore candidate immune–fibroblast communication pathways that may contribute to these responses.

Aim and Objectives

Aim

To characterize macrophage and fibroblast transcriptional states associated with fibrotic remodelling during the biomaterial-induced foreign body response and to explore candidate immune–stromal communication programs linked to these states.

Objectives

1. Establish a high-quality single-cell dataset from biomaterial implantation sites through rigorous quality control and preprocessing.

2. Define the immune and stromal cellular landscape of the foreign body response by annotating major cell populations.
3. Resolve functional macrophage and fibroblast states associated with inflammation, tissue remodelling, and fibrotic extracellular matrix production.
4. Identify candidate macrophage–fibroblast signalling interactions associated with fibrotic fibroblast programs using hypothesis-generating ligand–receptor analysis.

Methods

Single-cell RNA sequencing data were obtained from the Gene Expression Omnibus (GEO) under accession **GSE203099**, comprising cells isolated from biomaterial implantation sites (Kim et al., 2022). Standard quality control procedures were applied to exclude low-quality or stressed cells, followed by normalization and identification of highly variable genes. Unsupervised clustering was used to define transcriptionally distinct cell populations, which were annotated based on established canonical marker genes to identify major immune and stromal cell types.

Macrophage/monocyte and fibroblast populations were analysed in greater depth through subclustering to resolve transcriptional sub-states. To facilitate biological interpretation, curated gene programs related to inflammation, antigen presentation, phagocytic activity, tissue remodelling, and fibrotic extracellular matrix production were scored across subclusters. To explore potential immune–stromal interactions, a focused ligand–receptor analysis was performed based on transcriptional co-expression of biologically plausible signalling pairs, with the aim of generating hypotheses regarding macrophage-derived signals associated with fibroblast activation programs rather than inferring direct signalling activity.

Results

Immune and stromal landscape of biomaterial implantation

Single-cell profiling revealed a complex immune and stromal environment at biomaterial implantation sites (Figure 1). The cellular landscape was dominated by macrophages and monocytes, which together accounted for approximately 70% of all profiled cells, followed by fibroblasts (~12%), with additional contributions from neutrophils, endothelial cells, and adaptive immune populations. This composition is consistent with a classical foreign body

response, in which innate immune cells and stromal fibroblasts play central roles in sustaining inflammation and driving fibrotic remodelling.

Functional heterogeneity of macrophage states

Focusing on the macrophage compartment, subclustering and gene program scoring revealed substantial functional heterogeneity (Figure 2). Distinct macrophage populations were characterized by enrichment of inflammatory, interferon-associated, antigen presentation, phagocytic/lysosomal, and tissue remodelling–associated programs. Notably, a subset of macrophage states exhibited strong enrichment of pro-fibrotic–associated gene programs, including *Tgfb1*, *Spp1*, and *Fn1*, while displaying relatively low inflammatory signatures. This transcriptional profile suggests the presence of macrophage states potentially specialized for tissue remodelling rather than acute inflammatory responses.

Diversity of fibroblast activation programs

Analysis of the fibroblast compartment revealed multiple transcriptional programs associated with extracellular matrix production, myofibroblast features, and TGF-β responsiveness (Figure 3). Two fibroblast clusters showed markedly elevated ECM and TGF-β response scores and were designated as ECM-high fibroblasts, consistent with fibrotic activation. Other fibroblast populations displayed intermediate or low ECM enrichment, while a distinct cluster exhibited a myofibroblast-like transcriptional profile without strong ECM or TGF-β signatures, highlighting multiple axes of fibroblast activation within the foreign body response.

Candidate macrophage–fibroblast signalling interactions

Given the coexistence of pro-fibrotic–associated macrophage states and ECM-high fibroblast programs, we next examined candidate immune–stromal communication pathways (Figure 4). Ligand–receptor analysis identified distinct macrophage–fibroblast interaction patterns associated with fibrotic fibroblast states. Interactions involving pro-fibrotic–associated macrophages were preferentially enriched toward ECM-high fibroblasts, with **SPP1–CD44** emerging as the top-ranked interaction. Additional candidate pathways included chemokine- and growth factor–associated signalling axes, such as CCL3–CCR1, TGFB1–TGFBR1/2, and PDGFB–PDGFRB, suggesting a multifactorial immune signalling environment linked to fibrotic remodelling. These associations are hypothesis-generating and do not imply direct signalling activity.

Interpretation

Taken together, these findings indicate that biomaterial implantation is associated with distinct macrophage and fibroblast transcriptional states that may contribute to fibrotic remodelling. Notably, the association between macrophage functional state and fibroblast fibrotic programs suggests that immune cell activation profiles, rather than immune cell abundance alone, may play an important role in shaping fibrotic outcomes. Pro-fibrotic–associated macrophage states preferentially engaged ECM-high fibroblast programs through ligand–receptor pairs linked to matrix remodelling and fibroblast activation.

While the analyses presented here are inferential and do not establish causality, they are consistent with established roles of immune–stromal crosstalk in fibrosis and extend these concepts to the context of biomaterial-associated foreign body responses. These results provide a framework for linking immune cell transcriptional states to stromal remodelling programs and for prioritizing candidate signalling pathways for experimental validation.

Future work

Future studies could build upon these findings by integrating spatial transcriptomics to assess the physical proximity and organization of macrophage and fibroblast states at biomaterial interfaces. Complementary protein-level assays, such as cytokine profiling or immunostaining, would further clarify whether predicted ligand–receptor interactions correspond to active signalling pathways in vivo.

Beyond observational analyses, targeted perturbation experiments could directly test the functional relevance of candidate immune–fibroblast signalling axes. For example, CRISPR-mediated modulation of macrophage-expressed ligands such as *SPP1* or *TGFB1* could be used to assess downstream effects on fibroblast activation and ECM gene expression. In parallel, machine learning approaches integrating immune cell state composition and signalling features may enable prediction of fibrotic fibroblast enrichment across biomaterial conditions, providing a systematic framework for prioritizing immune–material interactions for further study.

Figures

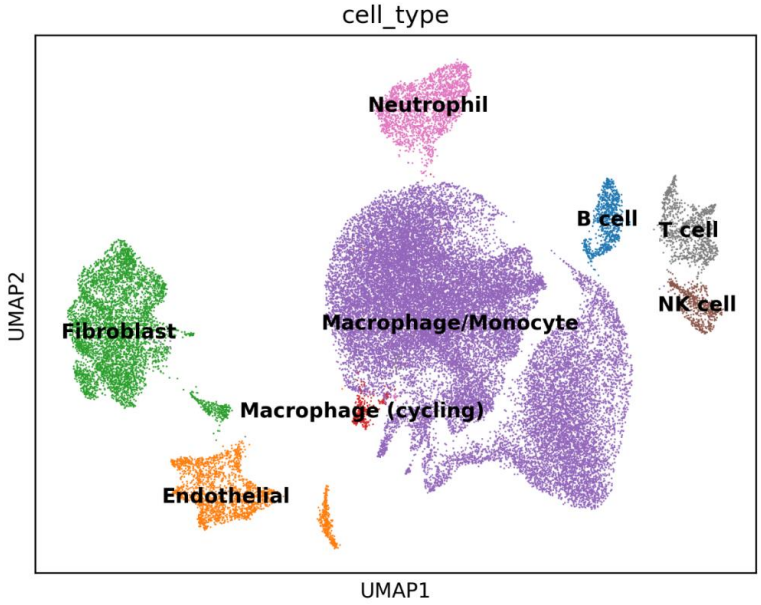

Figure 1. Immune and stromal landscape of biomaterial implantation

Single-cell profiling identifies a heterogeneous immune and stromal environment at biomaterial implantation sites, dominated by macrophages and monocytes, with additional contributions from fibroblasts, neutrophils, endothelial cells, and adaptive immune populations, consistent with a classical foreign body response.

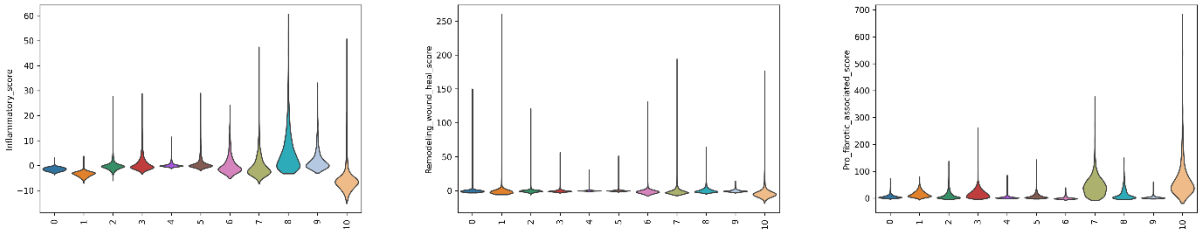

Figure 2. Functional heterogeneity of macrophage states

Violin plots of gene program scores across macrophage subclusters reveal distinct functional states spanning inflammatory, interferon-associated, antigen presentation, and tissue remodelling programs, including a subset enriched for pro-fibrotic–associated signatures.

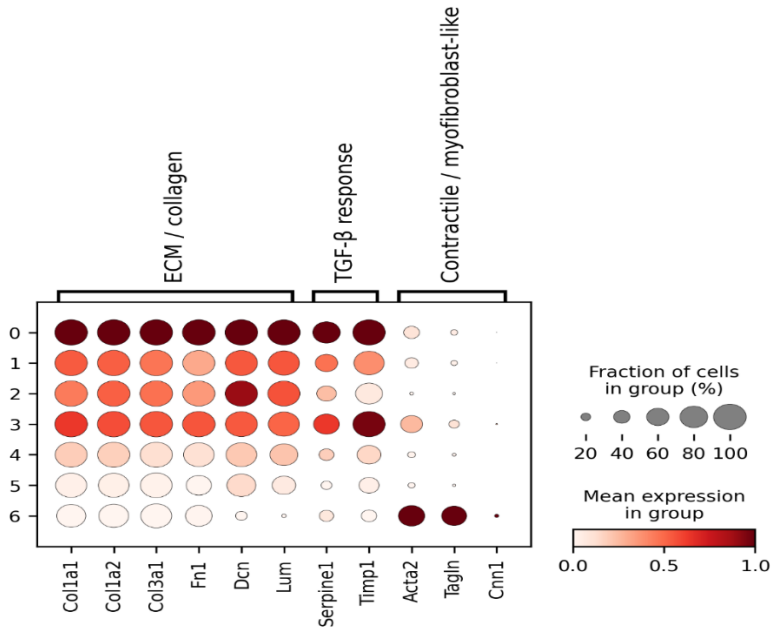

Figure 3. Diversity of fibroblast activation programs

Dotplot of fibroblast activation–associated genes across fibroblast subclusters identifies multiple fibroblast programs, including ECM- and TGF-β–responsive states consistent with fibrotic activation, alongside populations with intermediate or alternative transcriptional profiles.

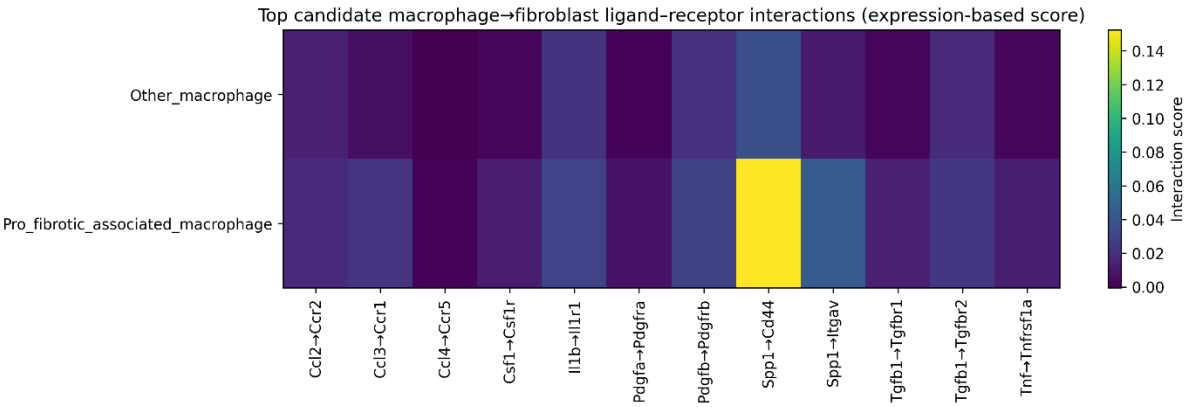

Figure 4. Candidate macrophage–fibroblast signalling interactions

Heatmap of inferred ligand–receptor interactions highlight candidate macrophage–fibroblast signalling patterns associated with fibrotic fibroblast programs, including SPP1–CD44 and chemokine- and growth factor–associated axes. Interactions are transcriptionally inferred and hypothesis-generating.

References

Kim, Y.-S., Shin, S., Choi, E. J., Moon, S. W., Jung, C. K., Chung, Y.-J., & Lee, S. H. (2022). Different molecular features of epithelioid and giant cells in foreign body reaction identified by single-cell RNA sequencing. *Journal of Investigative Dermatology, 142*(12), 3232–3242.e16. https://doi.org/10.1016/j.jid.2022.06.014

Liang, N. E., Parker, J., *et al.* (2024). Understanding the foreign body response via single-cell meta-analysis. *Biology, 13*(7), 540. https://doi.org/10.3390/biology13070540

Yang, B., Rutkowski, N., & Elisseeff, J. (2023). The foreign body response: Emerging cell types and considerations for targeted therapeutics. *Biomaterials Science, 11*(24), 7730–7747. https://doi.org/10.1039/D3BM00629H